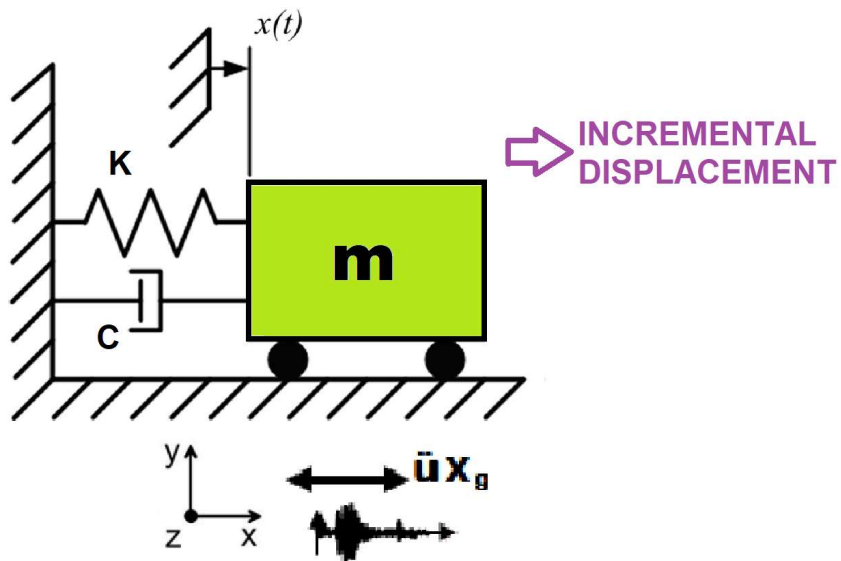


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL DAMPER STIFFNESS FROM ENERGY DISSIPATION CAPACITY INDEX USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES

WRITTEN BY SALAR DELAVAR GHASHGHAEE (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

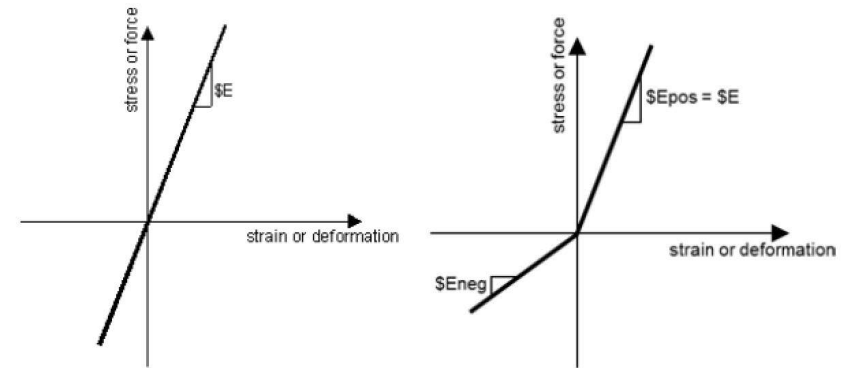
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

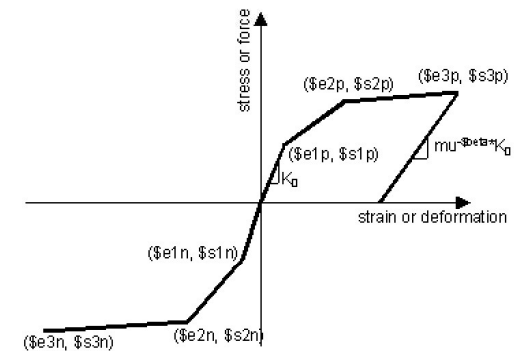
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

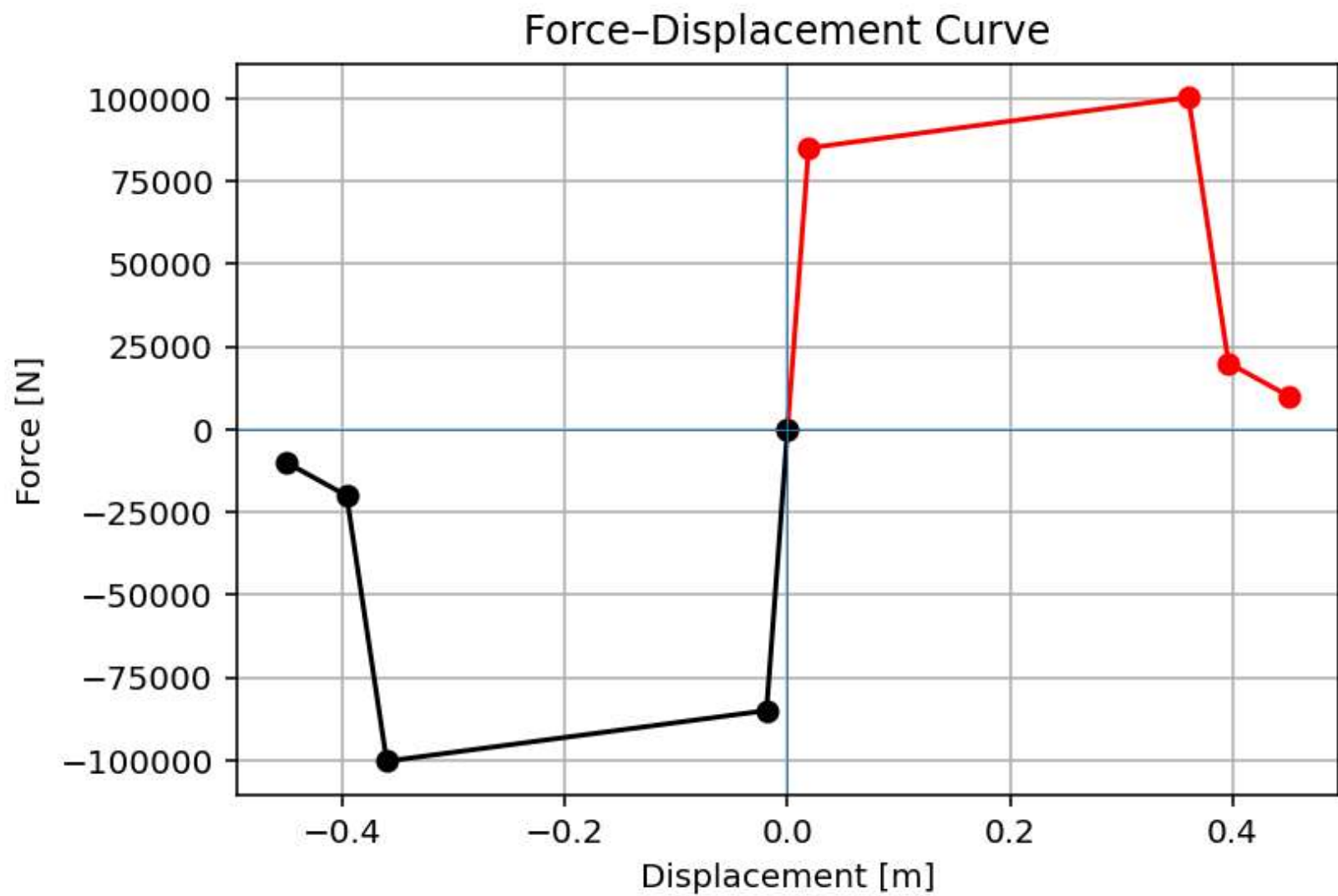
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL



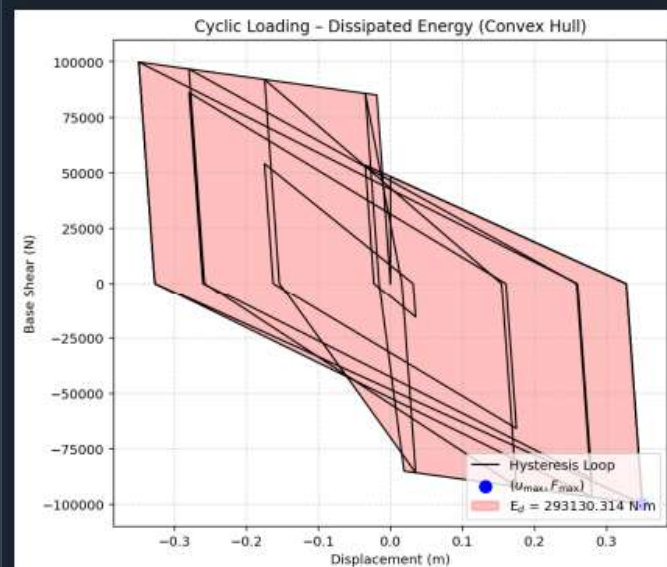
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K_Damper_OPTIMIZATION_DECI_SDOF.py X PARA_K_damp_OPTIMIZATION_DECI_SDOF.py X DISSIPATED_ENER

```
1 #####
2 # >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<
3 # DETERMINATION OF OPTIMUM STRUCTURAL DAMPER STIFFNESS FROM ENERGY DISSIPATION CAPACITY IND
4 # FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES
5 # -----
6 # COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STRUCTURE : AN OPENSE
7 # FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAMIC TIME-HISTORY AN
8 # -----
9 # ENERGY DISSIPATION CAPACITY INDEX = 100 * E_d(earthquake) / E_d(cyclic displacemen
10 # -----
11 # P(t) = P0 sin(wt)
12 # P(t) = P0 exp(-0.05wt) sin(wt)
13 # -----
14 # ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AND SYSTEMS AND EVAL
15 # OF ENERGY DISSIPATION CAPACITY INDEX
16 # -----
17 # THIS PYTHON SCRIPT IS WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)
18 # EMAIL: salar.d.ghashghaei@gmail.com
19 #####
20 """
21 Nonlinear Seismic Performance Assessment of a SDOF:
22 An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Static, Cyclic, and Ear
23
24 This OpenSeesPy script performs rigorous nonlinear static and dynamic analysis of a SDOF
25 for performance-based earthquake engineering. The 2D model incorporates both material nonlin
26 (elastic-perfectly plastic or hysteretic steel with strain hardening)
27 and geometric nonlinearity (corotational truss formulation) to capture P-delta effects
28 and large displacements-essential for collapse assessment.
29
30 Eight analysis protocols are implemented:
31 (1) [PERIOD] : Structural Period
32 (2) [STATIC] : Gravity load analysis establishing dead load state
33 (3) [PUSHOVER] : Displacement-controlled pushover generating full capacity curves
```

...PATION_CAPACITY_INDEX_SDOF\K_Damper_OPTIMIZATION_DECI_SDOF

43 %



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OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\Dell\Desktop\OPENSEES_FILES\+EIGHT_ANALY...TION_DECI_SDOF\K_Damper_OPTIMIZATION_DECI_SDOF.py

K_Damper_OPTIMIZATION_DECI_SDOF.py x PARA_K_damp_OPTIMIZATION_DECI_SDOF.py x DISSIPATED_ENER

```
1 #####
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```

Console 1/A x

```
+-----+
| lambda | omega | period | frequency |
+-----+
| 9.000e+02 | 30.0000 | 0.2094 | 4.7746 |
+-----+
```

Time: 20.0000, Displacement: 0.0033 mm, Reaction: -17.15 N

SEISMIC ANALYSIS DONE.

Xmax: 46669.67307237822
Dissipated Energy from Earthquake= 58626.06 N·m
Dissipated Energy from Cyclic Displacement= 293130.31 N·m

DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

ZONE 2: MINOR DAMAGE

Fmax: -6.93329127443576e-09
DF: -6.93326285272633e-06
DX: 4.099326685591892e-09
IT: 6 - RESIDUAL: 4.099326685591892e-09 X: 46669.67207237822

Optimum X (DAMPER STIFFNESS): 46669.6721
Iteration Counts: 6
Convergence Residual: 4.0993266856e-09

Total time (s): 67.1406

IPython Console Plots Variable Explorer Help Debugger History Files

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 920, Col 38 UTF-8 CRLF RW Mem 48%

Dissipated Energy from Earthquake= 58626.06 N·m

Dissipated Energy from Cyclic Displacement= 293130.31 N·m

DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

ZONE 2: MINOR DAMAGE

Fmax: -6.93329127443576e-09

DF: -6.93326285272633e-06

DX: 4.099326685591892e-09

IT: 6 - RESIDUAL: 4.099326685591892e-09 X: 46669.67207237822

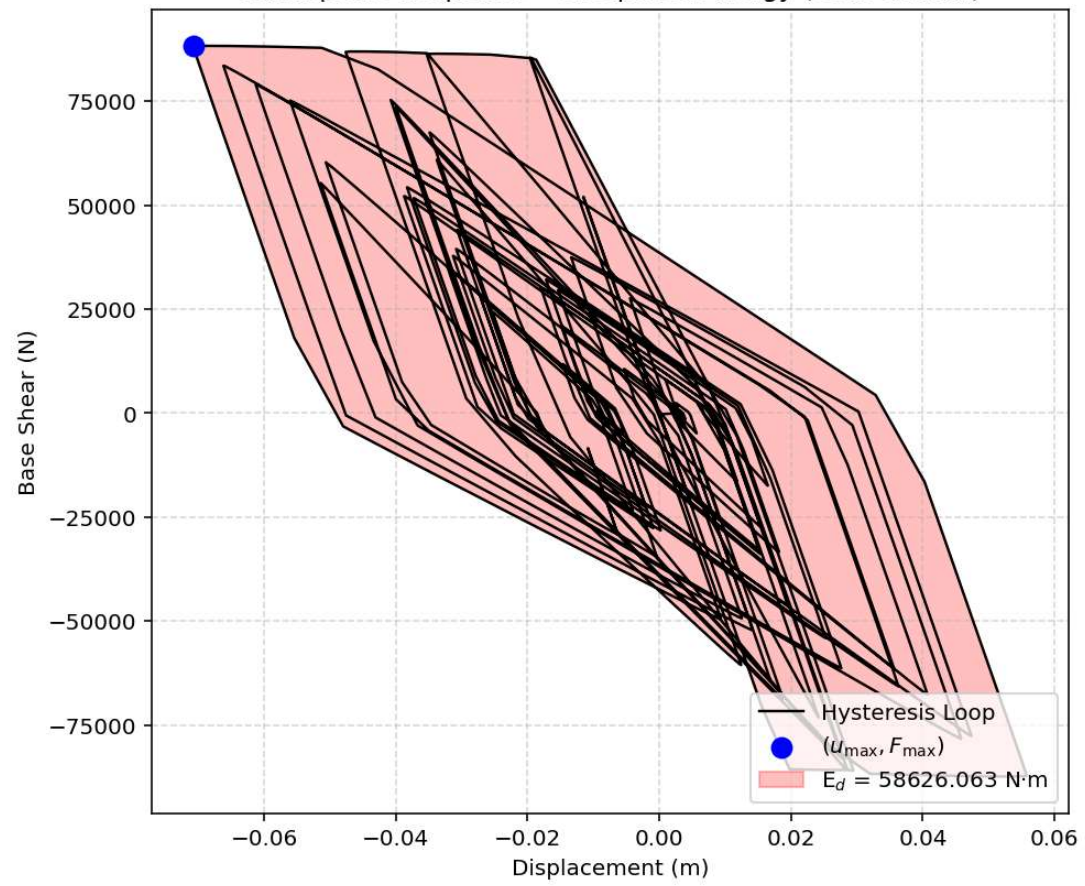
Optimum X (DAMPER STIFFNESS): 46669.6721

Iteration Counts: 6

Convergence Residual: 4.0993266856e-09

Total time (s): 67.1406

Earthquake Response - Dissipated Energy (Convex Hull)



Cyclic Loading - Dissipated Energy (Convex Hull)

